

NICHOLAS KO

Singapore | nicholas.ko.zy@outlook.com | nicholas-ko-zy.github.io

EXPERIENCE

National University of Singapore (NUS Cities)

Research Assistant

Singapore
Feb 2025 – Apr 2025

- Researched methods to project demographic changes at a national level, for cost-effective land-use planning.
- Refactored open-source projects (Python) to create synthetic populations from census data.

DSO NATIONAL LABORATORIES

Operations Research Analyst

Singapore
Aug 2024 – Feb 2025

- Worked across engineering teams and military users to develop decision-making models for highly classified military scenarios.
- Developed discrete event simulations using Python and Ansys Systems Toolkit (STK).

PROJECTS

Open Source Vehicle Routing Solver

May 2025 – Present

- Built a vehicle routing web application to solve and visualise the Vehicle Routing Problem (VRP).
- Achieved routing results for 30 locations and a fleet of 4 vehicles within ~5 seconds.
- Implemented using Django, Leaflet, Open Source Routing Machine (OSRM), Google OR-Tools, and Nominatim.
- Current work in progress: Scaling up the fleet size and order of visit per route.
- Link to repo: https://github.com/nicholas-ko-zy/django_map

Past Experience

ADATOS

Geospatial Analyst Intern

Singapore
May 2023 – Jul 2023

- Worked on crop yield prediction models (Python + QGIS) trained on satellite (multispectral/SAR) images of agricultural plots.
- Implemented new pipeline that reduced image segmentation time by over 50% for land-use land-cover mapping to track deforestation / verify carbon projects in South America.
- Provided financial projections in Excel/Python with worst-case, base-case, and best-case, for agriculture company's fundraising efforts.

YALE-NUS COLLEGE

Research Assistant

Singapore
May 2021 - Jul 2021

- Advisor: Assoc. Professor Michiel van Breugel.
- Tested if ground elevation and species abundance could predict mangrove diversity in research plots.
- Performed bootstrap resampling to create synthetic population of 5000 + mangrove trees on Pulau Ubin.
- Constructed visualisations in Rstudio (dplyr, ggplot2).

EDUCATION

YALE-NUS COLLEGE

B.S. Mathematical, Computational and Statistical Sciences (Data Science Track)

Singapore
2020 - 2024

- GPA: 4.35/5.00
- Coursework: Stochastic Processes, Graph Theory, Finite Geometry, Generalised Linear Models, Statistical Inference, Probability, Linear Algebra, Data Structures and Algorithms.

ADDITIONAL INFORMATION

- Languages: English (native), Cantonese (conversational)
- Programming Languages: Python, R
- Software: QGIS, Ansys Systems Tool Kit (STK)
- Github: <https://github.com/nicholas-ko-zy> | LinkedIn: <https://www.linkedin.com/in/nicholas-ko-zy>